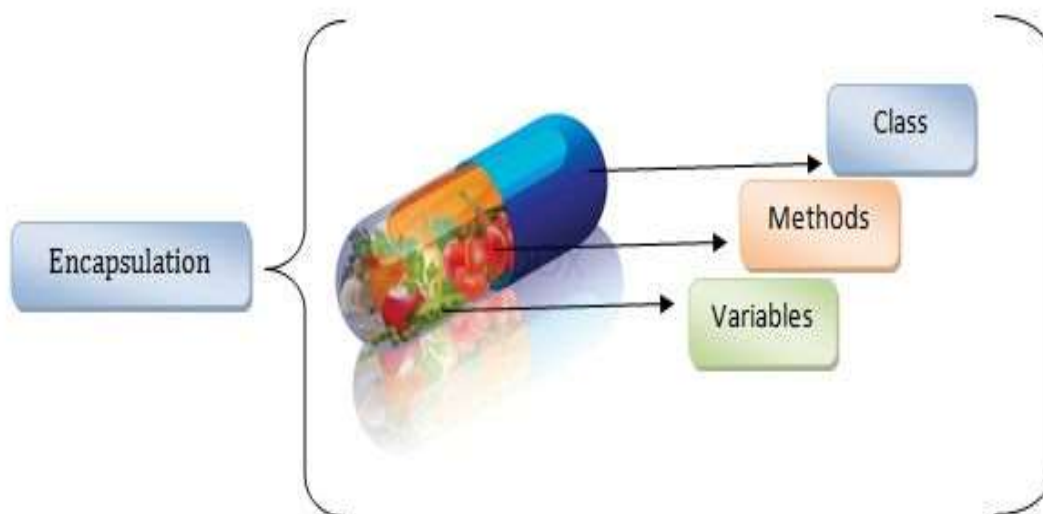


Encapsulation in Java

✚ What is Encapsulation :-

- Encapsulation in Java is one of the four pillars of OOP (Object-Oriented Programming).
- It is the process of **wrapping data (variables)** and **methods (functions)** together into a single unit (class).
- It also restricts direct access to the data and provides control through **getters and setters**.



✚ Key Points :-

- Achieved using **classes** and **access modifiers** (private, public, protected).
- Data members (variables) are kept **private**.
- Access to data is given through **public getter and setter methods**.
- Helps in **data hiding** and **security**.

✚ Advantages of Encapsulation :-

- **Data Hiding:** Prevents unauthorized access to fields.
- **Control:** Getters/Setters allow validation and controlled access.
- **Reusability:** Makes code easier to reuse and maintain.
- **Flexibility:** Internal implementation can change without affecting other classes.

✚ How to perform Encapsulation :-

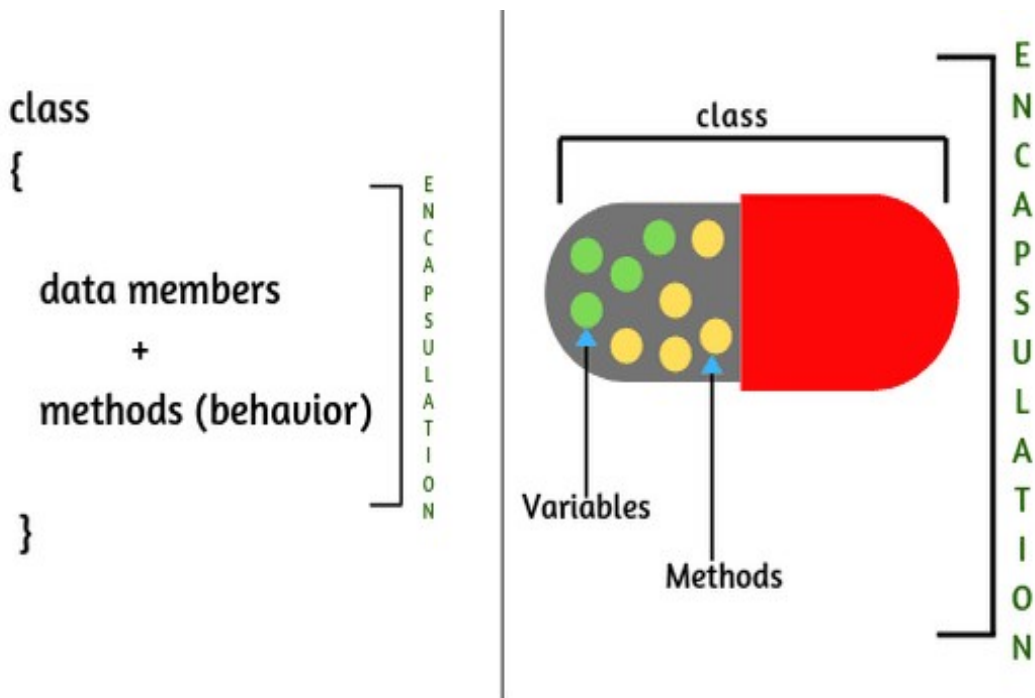


Fig: Encapsulation

Implementation of Encapsulation

Input :-

```
1 package Oops;
2
3 public class encapsulation {
4     private long student_aadhar_no;
5     private String student_account_number;
6
7     void set_s_acno(String student_account_number)
8     {
9         this.student_account_number=student_account_number;
10    }
11    String get_s_acno()
12    {
13        return student_account_number;
14    }
15
16    void set_aadharno(long student_aadhar_no)
17    {
18        this.student_aadhar_no=student_aadhar_no;
19    }
20    long get_aadharno()
21    {
22        return student_aadhar_no;
23    }
24
25    public static void main(String[] args) {
26        encapsulation ob=new encapsulation();
27        ob.set_aadharno(5157787310561);
28        System.out.println("Student aadhar number is : "+ob.get_aadharno());
29
30        ob.set_s_acno("SBI3456");
31        System.out.println("Student account number is : "+ob.get_s_acno());
32    }
33 }
```

Output :-

```
Student aadhar number is : 515273724256
Student account number is : SBI3456
```

Input :-

```
1 package Oops;
2
3 public class encapsulation_e {
4     private int employee_id;
5     private float employee_salary;
6
7     void set_e_id(int employee_id)
8     {
9         this.employee_id=employee_id;
10    }
11    int get_e_id()
12    {
13        return employee_id;
14    }
15
16    void set_e_salary(float employee_salary)
17    {
18        this.employee_salary=employee_salary;
19    }
20    float get_e_salary()
21    {
22        return employee_salary;
23    }
24
25    public static void main(String[] args) {
26        encapsulation_e ob=new encapsulation_e();
27        ob.set_e_id(101);
28        System.out.println("Employee id is : " +ob.get_e_id());
29
30        ob.set_e_salary(60000);
31        System.out.println("Employee salary is : " +ob.get_e_salary());
32    }
33 }
```

Output :-

```
Employee id is : 101
Employee salary is : 60000.0
```